

CSE-4203 - Number Theory & Cryptography

① Defⁿ 1: if $a, b \in \mathbb{Z}$ and $a \neq 0$, $c \in \mathbb{Z}$ such that $b = ac$
then we say "a divides b"
 b/a or $\underline{a} \mid \underline{b}$ | eg: $\begin{array}{r} 3 \overline{) 7} \\ 12 \overline{) 24} \end{array} \rightarrow \begin{array}{l} r=0 \\ q=2 \end{array}$

② Defⁿ 2: $\Rightarrow \begin{array}{c} \text{divisor} \\ (d) \end{array} \overline{\begin{array}{c} \text{dividend} \\ (a) \end{array}} \begin{array}{c} \text{quotient} \\ (q) \end{array}$
 $\underline{\hspace{1cm}}$
 $\begin{array}{c} \text{remainder} \\ (r) \end{array}$

$$a = d \cdot q + r$$

$$\rightarrow q = d \mid a = \lfloor a/d \rfloor$$

$$r = a \bmod d \rightarrow 0 \leq r < d$$

* Defn-3: if $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$ then
 $a \equiv b \pmod{m}$, if $\underline{m \mid (a-b)}$
 s/w $a \not\equiv b \pmod{m}$

eg: $100 \not\equiv 2 \pmod{11} \rightarrow m \nmid (a-b)$
 $\downarrow \quad \downarrow \quad \downarrow$
 $a \quad b \quad m$
 $\rightarrow 11 \nmid (100-2) \rightarrow 11 \nmid 98$

~~$101 \equiv 2 \pmod{11} \rightarrow 11 \mid (101-2) = 11 \mid 99$~~ $\rightarrow$ $a \% m$
 $b \% m$

* Theorem-3 if $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$ then
 $a \equiv b \pmod{m}$ iff $\underline{(a \bmod m) = (b \bmod m)}$

$a, b \rightarrow$ returns same remainder upon div by m

$$\begin{array}{r} 11 \overline{) 101} \quad 9 \\ \underline{99} \\ 2 \end{array}$$

$$\underline{101 \bmod 11 = 2}$$

$$\begin{array}{r} 11 \overline{) 2} \quad 0 \\ \underline{0} \\ 2 \end{array}$$

$$\underline{2 \bmod 11 = 2}$$

(*) Theorem - 4: Let $m \in \mathbb{Z}^+$ then a, b are congruent modulo
 m if $K \in \mathbb{Z}$ such that
 $\rightarrow a = b + Km$

Derivation

$$a \equiv b \pmod{m} =$$

$$\rightarrow m \mid (a-b)$$

$$(a-b) = m \cdot K$$

$$a = b + Km$$

(*) Theorem - 5: Let $m \in \mathbb{Z}^+$ and

$$\begin{cases} a \equiv b \pmod{m} \\ c \equiv d \pmod{m} \end{cases}$$

$$\rightarrow (1) \quad (a+c) \equiv (b+d) \pmod{m}$$

$$(2) \quad ac \equiv bd \pmod{m}$$

⑧ Corollary 2: if $m \in \mathbb{Z}^+$ and $a, b \in \mathbb{Z}$ then

① $(a+b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$

② $(ab) \bmod m = ((a \bmod m)(b \bmod m)) \bmod m$

⑧ Arithmetic modulo m $\xrightarrow{+}$ $\begin{matrix} 0 \\ \pm \end{matrix} \dots m$

if $\underline{z}_m \in \mathbb{Z}^+$, and $z \in \underline{z}_m < \underline{m}$ then

① Addition $\rightarrow a +_m b = (a+b) \bmod m$

② Multi $\rightarrow a \cdot_m b = (ab) \bmod m$

eg: $7 +_{11} 9 = (7+9) \bmod 11 = 16 \bmod 11 = 5$

$7 \cdot_{11} 9 = (7 \cdot 9) \bmod 11 = 63 \bmod 11 = 8$

4.2

Self study

⑤

Interconv.

①

Binary exp.

②

Octal "

③

Hex "

④

Base Conversion